

would be going from the root to A in the 1-combination level, then down to B in the 2-combination level, then to C in the 3-combination level, where we stop with our complete 3-combination: ABC. For the next combination, we traverse from the root to A to B to D, giving us the combination ABD. We continue doing this for ACD and BCD. Our tree has four nodes in the 3-combination level, and four 3-combinations from {A, B, C, D}: ABC, ABD, ACD, and BCD.

Notice that we create the tree in [Figure 6-4](#) by starting with a blank string for the root node. This is the 0-combination level, and it applies to all combinations of zero selections from the set; it's simply an empty string. The child nodes of the root are all elements from the set. In our case, that is all four elements from {A, B, C, D}. While sets don't have an order, we need to be consistent in using the ABCD order of the set while generating this tree. This is because every node's children consist of the letters after it in the ABCD string: all A nodes have children B, C, and D; all B nodes have children C and D; all C nodes have one D child; and all D nodes have no child nodes.

While it's not directly related to the recursive combination function, also notice the pattern in the number of k -combinations at each level:

- The 0-combination and 4-combination levels both have one combination: the empty string and ABCD, respectively.
- The 1-combination and 3-combination levels both have four combinations: A, B, C, D and ABC, ABD, ACD, BCD, respectively.
- The 2-combination level in the middle has the most combinations at six: AB, AC, AD, BC, BD, and CD.

The reason the number of combinations increases, peaks in the middle, and then decreases is that the k -combinations are mirrors of each other. For example, the 1-combinations are made from the elements not selected for the 3-combinations:

- The 1-combination A is the mirror of the 3-combination BCD.
- The 1-combination B is the mirror of the 3-combination ACD.
- The 1-combination C is the mirror of the 3-combination ABD.